BROOKFIELD TEXTURE PRO CT®

Application Software

Operating Instructions

Manual No. **M08-373**

BROOKFIELD

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I. INTRODUCTION

I.1. About TexturePro CT Software

TexturePro CT is designed for use with the CT3 Texture Analyzer and a Windows 2000 or above operating system. It collects data from the texture analyzer and allows it to be saved, viewed, printed, plotted, and analyzed. Features include:

- Creation of new test methods within a single window.
- Entering of sample description and information in a single window.
- Optional Secure mode, in which an operator can run test methods and collect and save new data, but existing test methods and data are protected against change or deletion.
- Results screen that is test-method intelligent, allowing selection of only relevant results based upon the type of test performed.
- Statistical analysis of up to 30 samples.
- True database file storage structure, which minimizes space requirements and allows for easy, logical file naming conventions.
- Easy export of data files to other computers or into Microsoft Excel format.
- Custom page, in which the operator can configure their own, customized display.

I.2. System Requirements

Minimum System Requirements	
Microprocessor	800MHz Pentium processor (or equivalent)
Memory	512 MB of RAM
Hard Drive Space	30 MB available
Video	VGA (800 x 600 resolution)
Operating System:	MS-Windows 2000, XP, Vista
Communications Port:	One USB or RS-232 Port

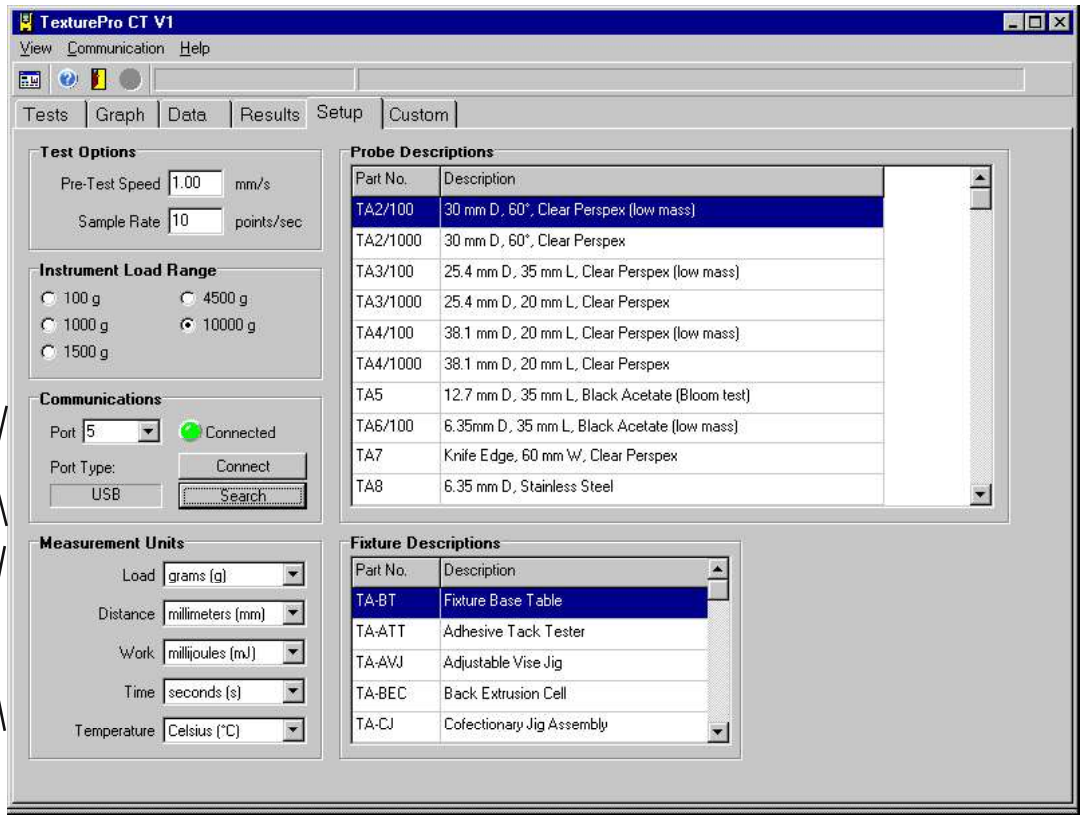
II. GETTING STARTED

Before running TexturePro CT for the first time, the following steps should be taken:

- Establish Communications
- Choose the Measurement Units

II.1 Establishing Communications

The communications control window in the set up page Setup is used to establish communications with the instrument. These same features can be accessed from the Communication in the main menu.



Establishing Communications for the First Time:

USB

1. Install the software first.
2. Connect the USB cable to the computer and the instrument.
3. Turn on the instrument. Windows should display several messages as it automatically detects a new USB device and locates the driver for it. When it is done, Windows should indicate that the device/hardware is installed and ready to use.

4. Restart the instrument and set it to the Remote mode.
5. Start the software. Once the software is up and running, go to the Setup page and click the "Search" button. The software should find the port which the instrument is connected to and reset the instrument.

RS-232

1. Install the software first.
2. Connect the RS-232 cable to the computer and the instrument.
3. Turn on the instrument and set it to the Remote mode.
4. Start the software. Once the software is up and running, go to the Setup page and click the "Search" button. The software should find the port which the instrument is connected to and reset the instrument.

Establishing Communications on a Regular Basis

- The easiest way to ensure communications is to have the instrument connected and turned on before starting the software. The software will then automatically connect with the instrument at start up.
- If you turn on the instrument after the software has started, hit the "Connect" button to start communications.
- If you change COM ports, hit the "Search" button to find the new port and start communications.

II.2 Choosing the Measurement Units

Use the Measurement Unit drop down lists to choose the desired display units for various parameters.

Certain measurement units are determined by other units, or are fixed. Speed units will be the chosen distance units divided by the chosen time units. For example, if centimeters are selected for distance units and minutes for time units, speed will be in centimeters per minute (cm/min). Apparent Modulus and Peak Stress values are always in dynes per square centimeter (dyn/cm^2), and the Sample Rate is always in points per second (points/sec).

NOTE: Each time a unit is changed, the software must be restarted.

III. RUNNING A TEST

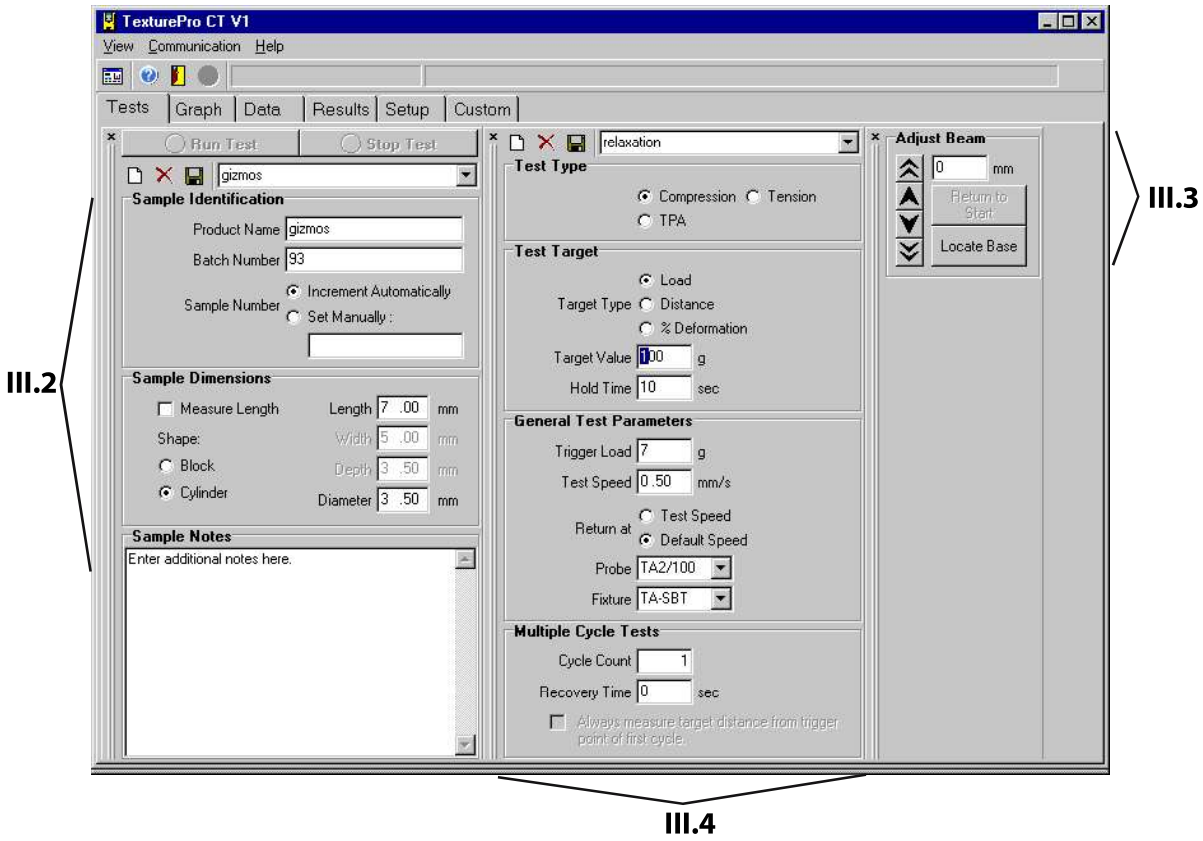
Section 3 describes the features and functions of each TexturePro CT page and dialogue box.

Before a sample can be tested, the following things must be done:

- Communication must be established between the instrument and the software.
- The sample must be identified.
- The test method must be defined.

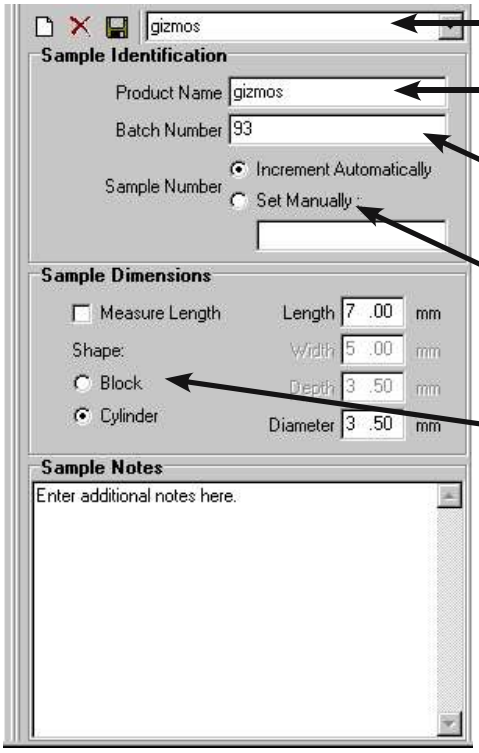
III.1 Test Page

Use the Tests page to define the sample to be tested and the test method to use.



III.2 Describing the Sample

Use this control window to describe the sample to be tested.



Lists all sample templates currently in the database.

Enter up to 30 alphanumeric characters to describe the product name of the sample.

Enter up to 30 alphanumeric characters to describe the batch number of the sample.

Enter up to 10 alphanumeric characters to describe the sample number of the sample, or have the software automatically choose the next sequential sample number.

Enter the sample dimensions here. If the "Block" shape is chosen, "Width" and "Depth" are used. If the "Cylinder" shape is chosen, "Diameter" is used.

III.3 Measuring the Sample Length

To have the instrument measure the sample length, follow these steps:

1. Click the "Locate Base" button in the "Adjust Beam" control window and follow the instructions.
2. Check the "Measure Length" checkbox in the "Sample Information" control window.

At the beginning of the next test, the instrument will measure the sample length. This is the length which will be recorded in the data file, and used for %deformation and strain calculations. Any value manually entered into the "Length" edit box will be ignored.

TexturePro CT can calculate the cross-sectional area of a sample based on two shape types: Block and Cylinder. The cross-sectional area is needed for the calculation of Stress results.

III.4 Defining the Test Method

Use this control window to define the test method to be used. Note: If the software is in Secure Mode, new test methods cannot be defined. However, existing test methods can be used. Choosing the Test Type

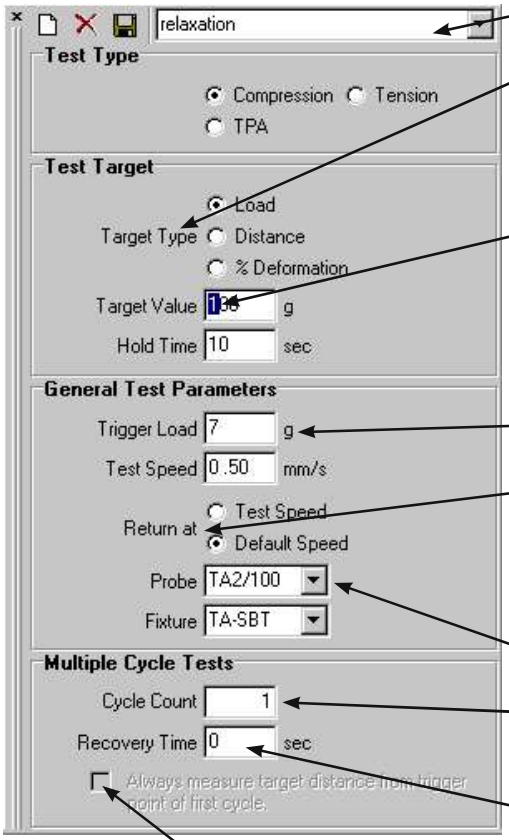
The CT3 Texture Analyzer can run Compression, TPA, or Tension tests.

In a Compression test, the sample is placed between the probe and the bottom fixture, and the probe moves down, pressing on the sample. The resulting data can be used for calculations such as Hardness and Fractures.

The TPA test is a specialized Compression test in which certain options have been set to conform with the texture profile analysis (TPA) method. Some calculations (such as Cohesiveness) are only valid if the TPA test method is used.

In a Tension test, the sample is clamped into the bottom and top fixtures, and the top fixture moves up, pulling the sample. The resulting data can then be used for various Tension calculations.

If a specific calculation is desired, it is best to review the testing requirements for that calculation (listed on the Results Page) before collecting the data. Calculations, which are not valid for the test method used, will be disabled.



List all test templates currently in the database..

Specify whether the probe will travel to a target load or a target ditance. Note: In order to see the % Deformation Target type, the sample length must be entered, or check "Measure Length" in sample dimension window. ***Note:*** The target type for Tension tests is always distance.

Specify the value of the target (load or distance) for probe travel.

Enter the amount of time the instrument will pause once it has reached the test target. Note: Hold time does not apply for TPA tests.

Enter the speed, which the probe will travel during test.

For a Compression test, the Return Speed determines how quickly the probe will move up after the compression stroke is done. Note: The return speed of a TPA test and a cycle test is fixed at the test speed. Return speed does not apply for Tension tests.

Indicates the probe and fixture which will be used for this test.

The number of times the test will repeat. Note: Cycle count is automatically set to 2 for a TPA test, and 1 for a Tension test.

Enter the amount of time the instrument will pause between test cycles. Note: Recovery time does not apply to Tension tests.

For Compression tests, when this option is chosen the target distance will always be measured from the location of the trigger point of the first cycle. Otherwise, the target distance is measured from the location of the new trigger point of each cycle. Note: This option is fixed as true for TPA test. This option does not apply for Tension tests or when the Target type is Load.

III.5 Secure Mode

NOTE: Secure Mode only applies if the software was installed using the "Secure" setup option.

When the software is in “Secure Mode”, it is indicated in the title bar at the top of the program. Secure Mode occurs when the software is started without using the Administrative password, which was set up for it. While in Secure Mode, the user can load and run existing test methods, and display and analyze data, but the following editing functions are unavailable:

- Saved data cannot be deleted or overwritten. The user can, however, choose not to save newly collected data.
- New test methods cannot be created.
- Sample, test, and report templates cannot be deleted.

In this way, the Secure Mode can be used to give some users limited access to the software.

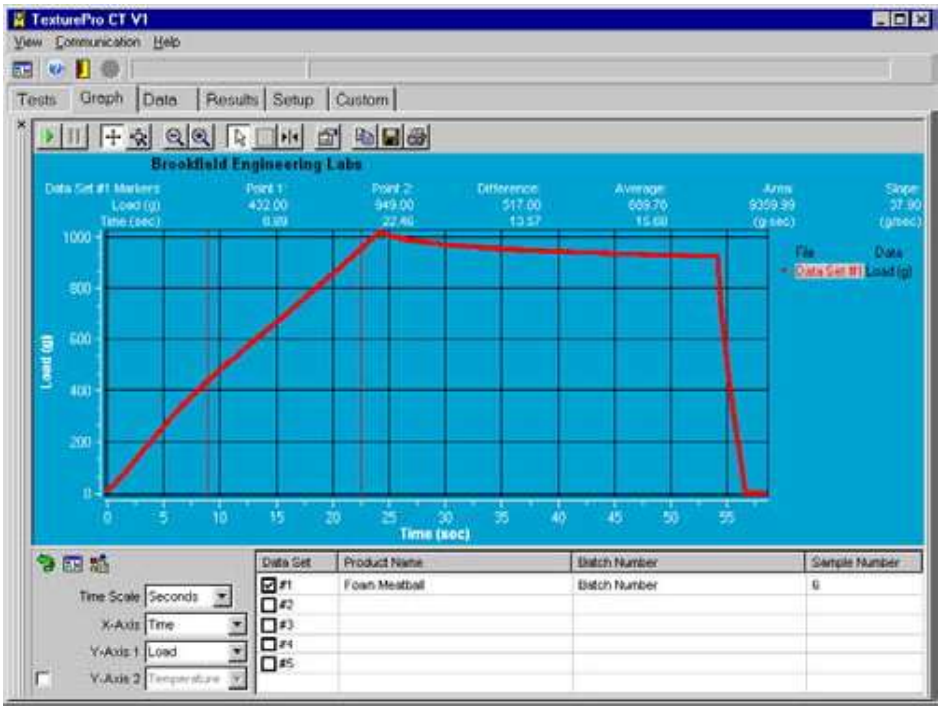
To set up an Administrative Password:

1. The Administrative password must be set up during the installation of the software. If a password was not set up during installation and you wish to set one up now, the software can be reinstalled.
2. During the installation, when asked which type of setup you wish to install, choose “Secure”, then click “Next”. You will be prompted to enter a password. Make careful note of the password for future reference.
3. Each time the software is started, the user will be asked to enter the password. If the password is entered, the software will start with full editing capabilities. If the password is not entered, the software will start in Secure Mode.
4. If the Administrative password is forgotten, the software must be reinstalled and a new password set up.
5. If you wish to remove the Secure Mode feature completely, the software must be reinstalled using the “Standard” setup type.

IV. GRAPHING

IV.1 Graph Page

During a test, data is graphed here as it is collected. In addition, up to five saved data files can be graphed together for comparison.



IV.2 Graph Options



Turns on tracking. As data is collected, axes will scroll to keep the newest data in view.

Turns off tracking.

Axis will scroll when the user drags it.

Axis will zoom when the user drags it.

Zooms both axes out.

Zoom both axes in.

 Allows a data point to be selected.

 Allows the user to draw a box, to which the graph will zoom.

 Places crosshairs with coordinates on the graph.

 Opens a dialog box with many graph display options.

 Copies the graph to the clipboard.

 Saves the graph as a bitmap.

 Prints the graph.

IV.3 Data Markers

Data Set #1 Markers:	Point 1:	Point 2:	Difference:	Average:	Area:	Slope:
Load (g)	432.00	949.00	517.00	689.76	9359.99	37.90
Time (sec)	8.89	22.46	13.57	15.68	(g·sec)	(g/sec)

Data markers can be used to inspect two individual points on the graph, and to show the relationship between them.

To place data markers on the graph:

1. Make sure the “Select” button on the Graph Options bar is depressed.
2. Move the mouse over to the desired point.
3. When the cursor changes to a pointing hand, click the point.
4. Repeat steps 2 and 3 to select a second point.
5. Cursors can be moved individually by dragging.
6. To remove a data marker, move the mouse over the marker and double-click.

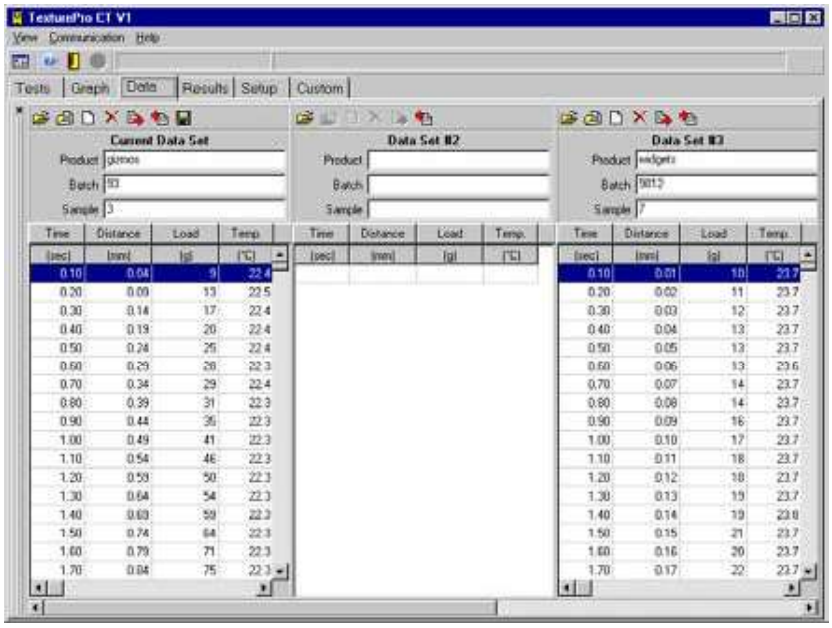
Once two points are selected, the following information is displayed at the top of the graph:

- The X and Y axis values of the points;
- The difference between the X and Y values of the two points;
- The average X and Y values of all points between the two selected;
- The area under the curve between the two points;
- The least-squares-fit slope of all the points between the two selected.

V. DATA

V.1 Data Page

The Data page can be used to display up to five data files, side by side. In its default configuration (set up for 800 x 600 pixel screen resolution), three of the data sets can be seen at once, and the other two are visible by using the scroll bar at the very bottom of the Data page. With higher resolutions, screens (1280 pixel width or higher) the Data control windows can be resized to view all five at once.




Open data file


Show sample details


Clear data grid


Delete data

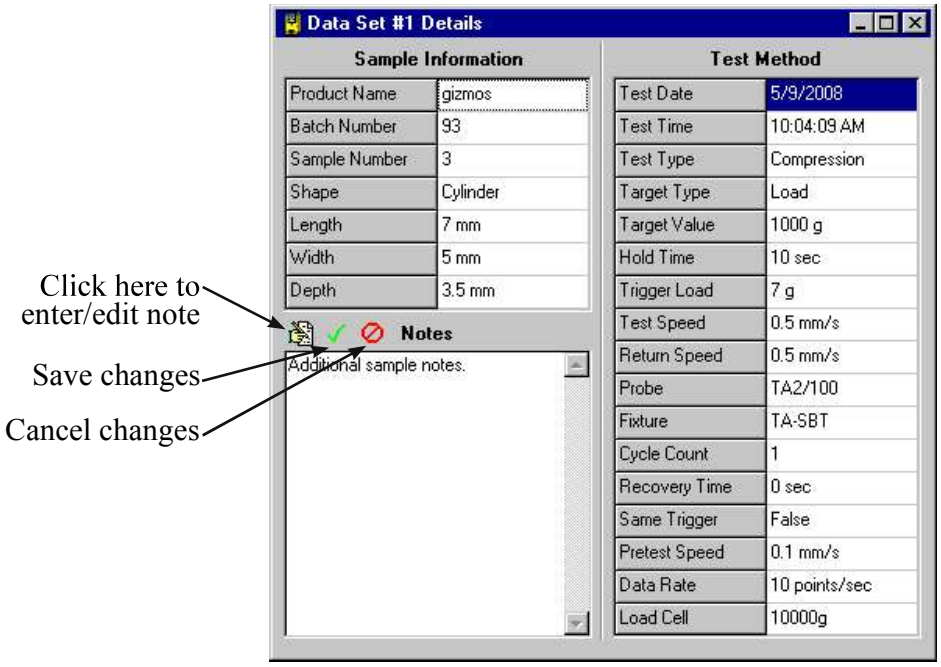

Export data


Import data


Save durrent data

V.2 Viewing Data Details

Clicking the Details button on the Data Page displays the sample description and test method information for the corresponding data file.



V.3 Moving Data Files

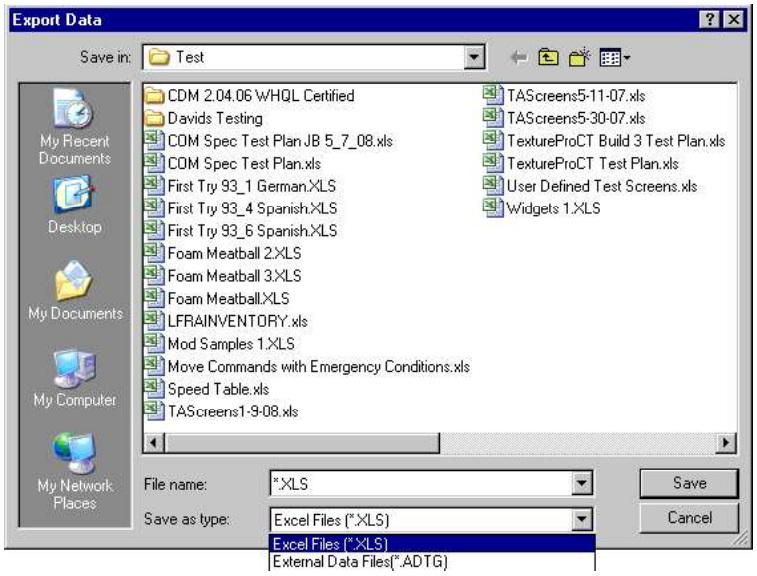
TexturePro CT data is stored in a compact form in a Microsoft Access database. Within a computer, all data files in the database are available through the software's Open Data dialog.

To transfer data files from one computer to another:

1. Export the data file as an External file (*.ADTG).
2. Move the resulting External file to the other computer and import the file using that computer's TexturePro CT.

V.4 Export Data

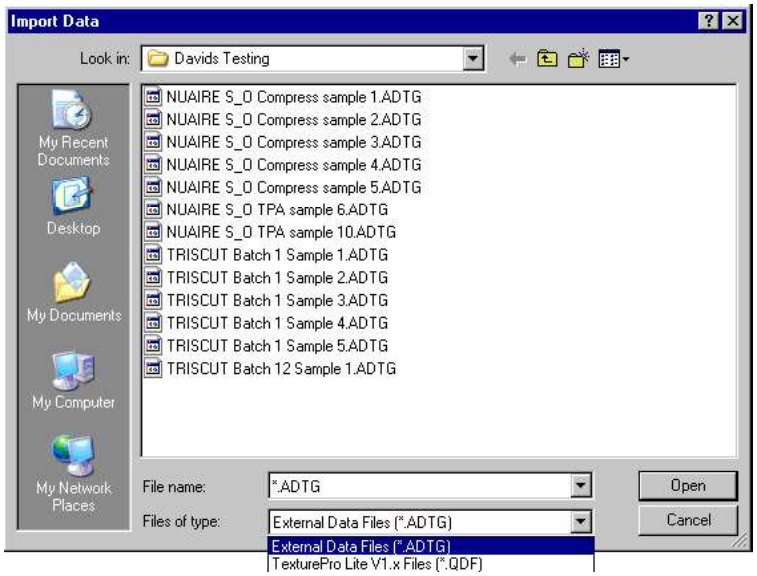
Data files can be exported to either an Excel format or an External file format. The External file format can be used to move data files from one computer to another, or to archive old data.



V.5 Import Data

Data files previously exported to the External file format, and legacy files, which were collected, using TexturePro Lite, can be imported into TexturePro CT.

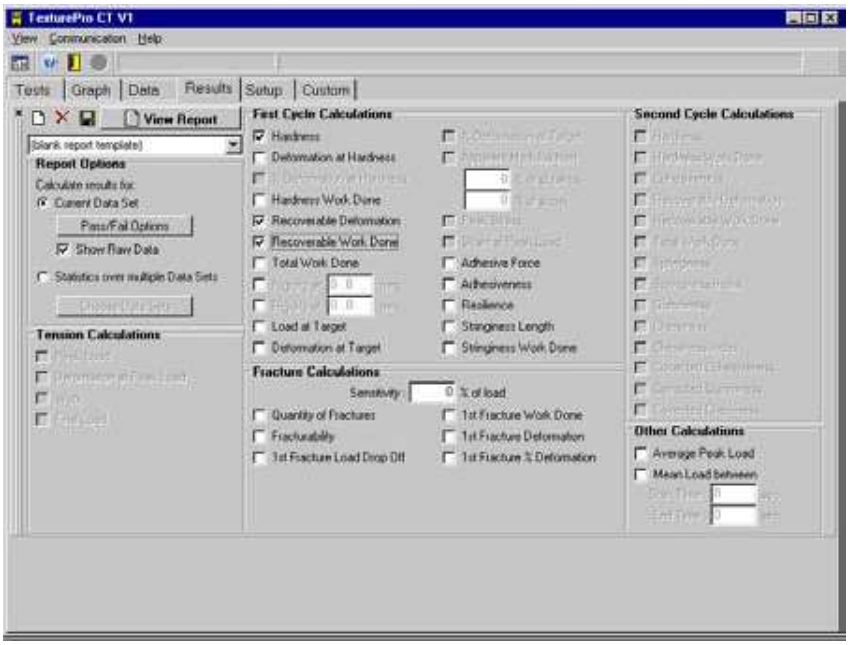
If a data set with the same Product Name / Batch Number / Sample Number already exists in the database, the user is asked if they want to overwrite the existing data. Note: If the software is in Secure Mode, imported data is not allowed to overwrite existing data.



VI . RESULTS

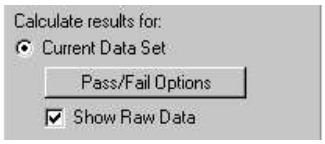
VI.1 Results Page

Use this page to analyze the current data set, or to perform statistical analysis on multiple data sets.



VI.2 Calculations on a Single File

Use the “Current Data Set” choice to calculate various results on a single data file, compare those results to specified pass/fail limits, and display those results along with the individual data points of the file.



To Calculate Results for a Single Data File:

1. Make sure there is data in the “Current Data Set” table on the Data page.
2. Click the “Current Data Set” calculation choice.
3. Check the various results to be calculated.
4. To set any pass/fail limits, click the “Pass/Fail Options” button and enter the result names and their limits in the “Pass/Fail Criteria” table.
5. To show the individual data points as well as the analysis results in the report, check the “Show Raw Data” checkbox.
6. Click the “View Report” button to view the report.

VI.3 Statistics over Multiple Files

Use the "Statistics over multiple Data Sets" choice to calculate results for several data files and display the individual results along with their minimum, maximum, average and standard deviation.

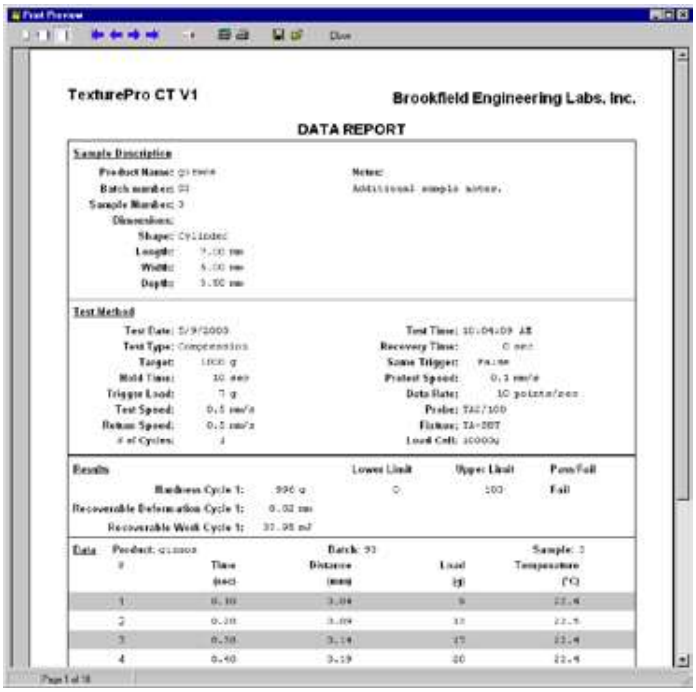


To Calculate Statistical Results over Several Data Files:

1. Click the "Statistics over multiple Data Sets" calculation choice.
2. Click the "Choose Data Sets" button and list up to 30 files to be analyzed in the "List Data Files" table.
3. Check the various results to be calculated. Statistics can be calculated on up to four results at one time.
4. Click the "View Report" button to view the report. Will only diddsplay four results at one time.

VI.4 Data Reports

When the "View Report" button is clicked, one of two reports is displayed. If the "Current Data Set" option is selected the "Data Report" is shown. If the "Statistics over multiple Data Sets" option is selected the "Statistical Report" is shown.



VI.5 Hardness Calculations

Hardness

- Sensory Definition: Maximum force required to compress a food between the molars.
- Mathematical Definition: Maximum load value of the compression cycle.
- Test Requirements: TPA or Compression test type.

Deformation at Hardness

- Mathematical Definition: Distance value at the Hardness point.
- Test Requirements: TPA or Compression test type.

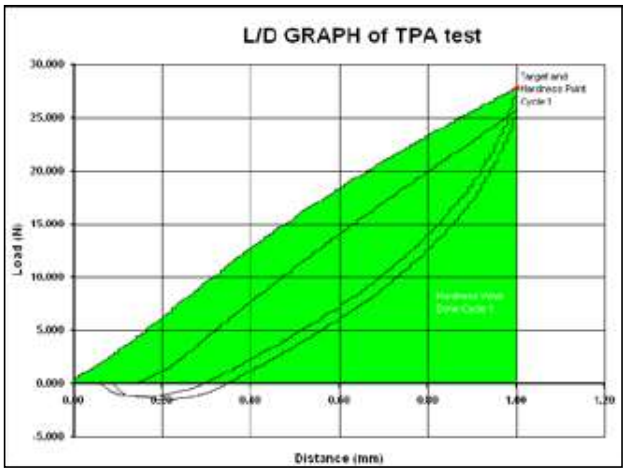
%Deformation at Hardness

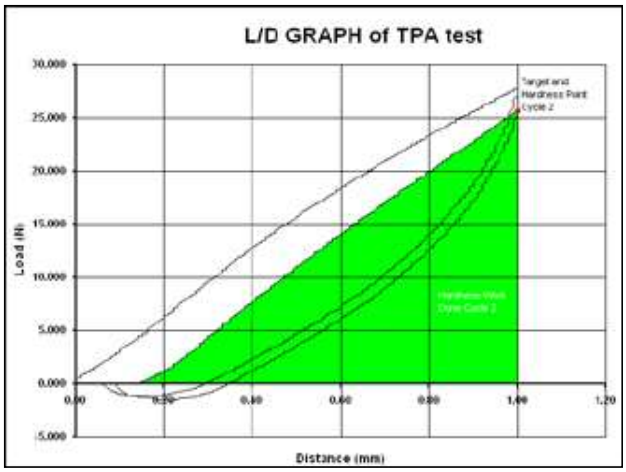
- Mathematical Definition: Deformation at Hardness / sample length * 100%.
- Test Requirements: TPA or Compression test type; Sample length must be defined.

Hardness Work Done

- Sensory Definition: Work necessary to overcome the internal strength of bonds within a food.
- Mathematical Definition: Area under the Load vs. Distance curve from the cycle start to the target value (load or distance). Note: Hardness Work Done is typically calculated for non-destructive tests. If a major fracture occurs before a target distance is reached, the Hardness Work Done will be calculated to the target distance, even though the Hardness point may occur at the fracture.
- Test Requirements: TPA or Compression test type.

Below are typical Load vs. Distance graphs showing these calculations.





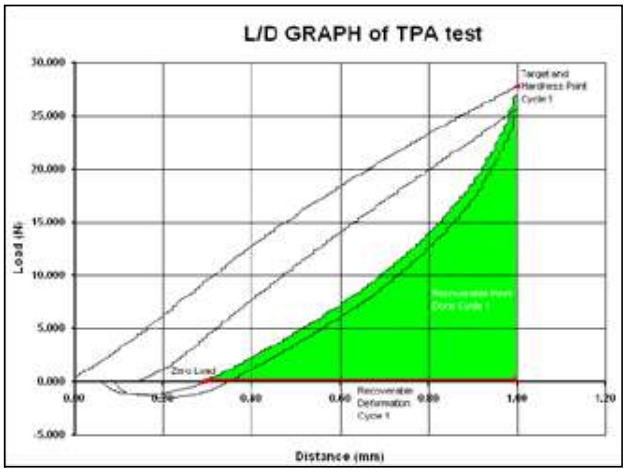
VI.6 Recoverable Calculations

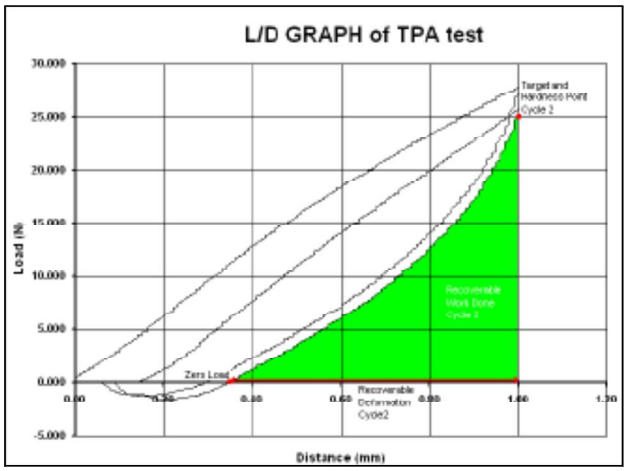
Recoverable Deformation

- Sensory Definition: Height recovered by a food upon removal of a compressing force.
- Mathematical Definition: Distance from the target (load or distance) down to zero load.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

Recoverable Work Done

- Sensory Definition: Work performed by the food against the compressing force as it is being removed.
- Mathematical Definition: Area under the Load vs. Distance curve, from the target (load or distance) down to zero load.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.





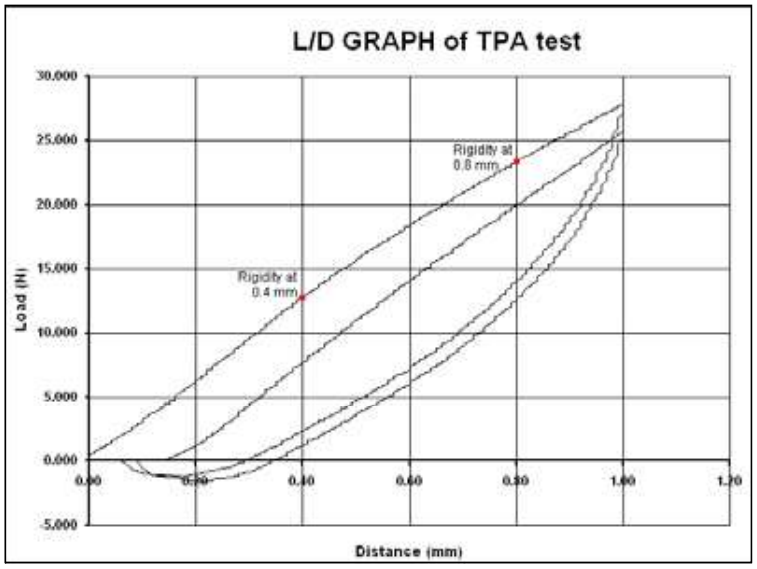
Total Work Done

- Mathematical Definition: Hardness Work Done + Recoverable Work Done.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

Rigidity

- Mathematical Definition: The load value at a specified distance during the first cycle compression stroke.
- Test Requirements: TPA or Compression test type.

Below is a typical Force vs. Distance graph showing these calculations.



VI.7 Calculations at Target

Load at Target

- Mathematical Definition: Load value at the target (load or distance). If a target type of load was chosen during testing, this value will be the same as the Hardness. If a deformation target type was chosen, this value may differ from the Hardness if there were major fractures during testing.
- Test Requirements: TPA or Compression test type.

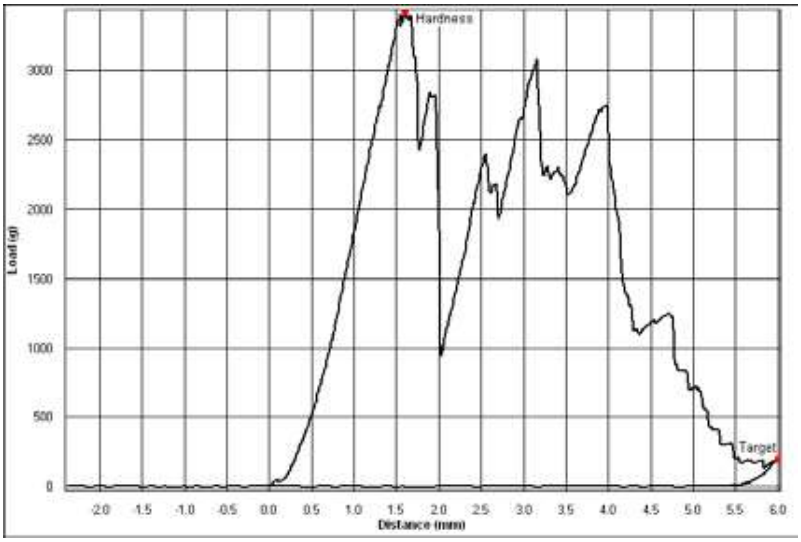
Deformation at Target

- Mathematical Definition: Distance value at the target (load or distance).
- Test Requirements: TPA or Compression test type.

%Deformation at Target

- Mathematical Definition: Deformation at Target / sample length * 100%.
- Test Requirements: TPA or Compression test type; Sample length must be defined.

Below is a typical Load vs. Distance graph showing these calculations.



VI.8 Stress/Strain Calculations

Apparent Modulus

- Sensory Definition: Gives an indication of the stiffness of a food.
- Mathematical Definition: Slope of a linear portion of the Stress vs. Strain curve during the first compression stroke. The slope is calculated between given limits, which represent percents of the load at target if there is no fracture, and percents of the fracture load if there is a fracture.
- Test Requirements: TPA or Compression test type; all sample dimensions must be defined.

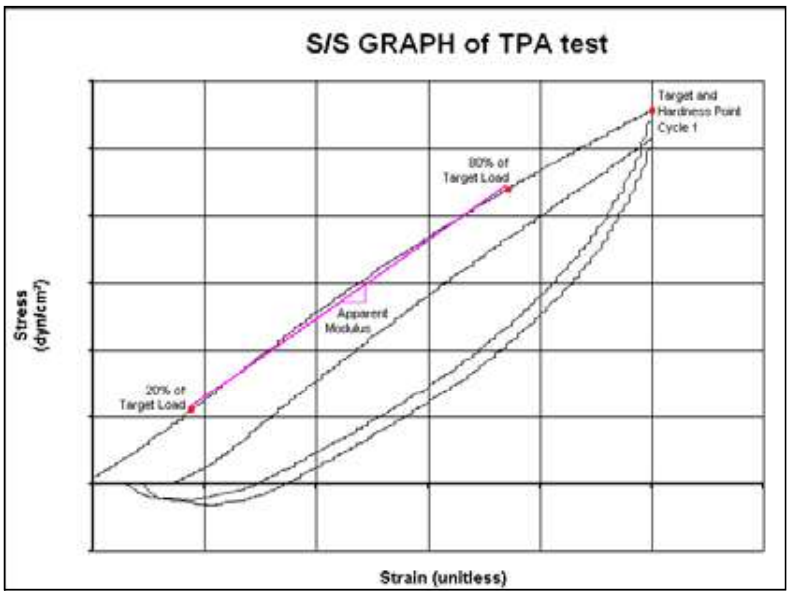
Peak Stress

- Mathematical Definition: The stress at the Hardness point of the first compression stroke. Stress is calculated as the load at a given point divided by the sample's cross-sectional area.
- Test Requirements: TPA or Compression test type; Sample width and depth must be defined.

Strain at Peak Load

- Mathematical Definition: The strain at the Hardness point of the first compression stroke. Strain is calculated as the deformation at a given point divides by the sample length.
- Test Requirements: TPA or Compression test type; Sample length must be defined.

Below is a typical Stress vs. Strain graph showing these calculations.



VI.9 Adhesive Calculations

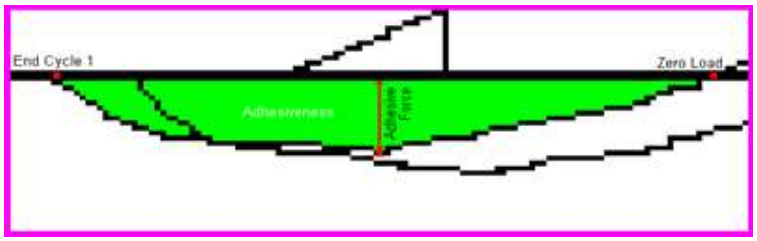
Adhesive Force

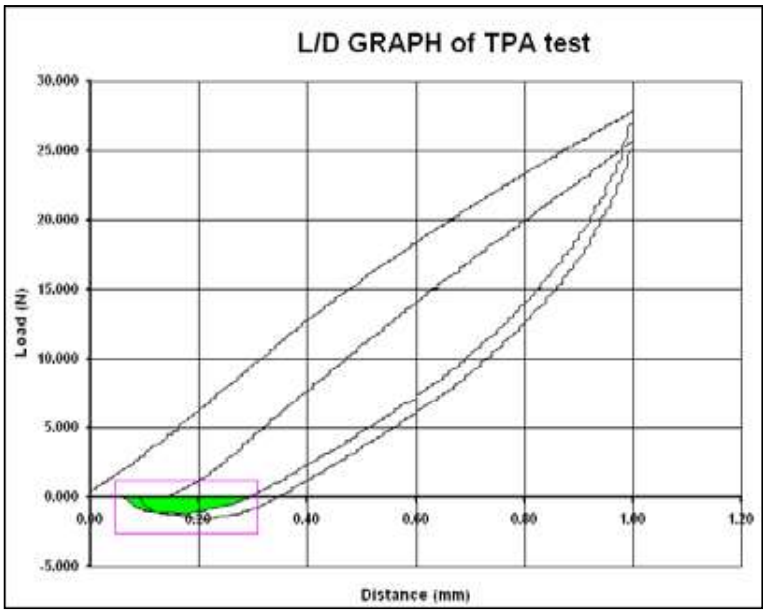
- Sensory Definition: The force necessary to overcome the attractive forces between the surface of the food and the surface of other materials, which the food comes into contact with (e.g. tongue, teeth, palate).
- Mathematical Definition: Maximum negative force of the first compression cycle.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

Adhesiveness

- Sensory Definition: Work required to pull food away from those surfaces.
- Mathematical Definition: Area under the Load vs. Distance curve, measured from where cycle 1 first reaches zero load to where it ends.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

Below are typical Force vs. Distance graphs showing these calculations.





Resilience

- Sensory Definition: Measurement of how a food recovers from deformation in relation to speed and forces applied.
- Mathematical Definition: Recoverable Work Done / Hardness Work Done, giving a value between 0 and 1.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

VI.10 Stringiness Calculations

Stringiness Length

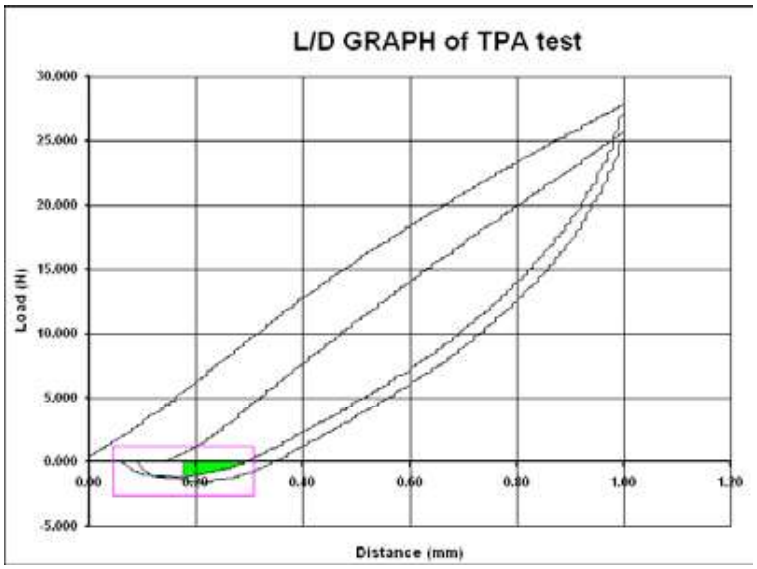
- Sensory Definition: The distance a food is extended as it is pulled away from a contact surface (e.g. teeth, palate).
- Mathematical Definition: Distance from the zero load of cycle 1 to the Adhesive Force.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

Stringiness Work Done

- Sensory Definition: Amount of work exhibited by the food as it clings to the contact surface.
- Mathematical Definition: Area under the Load vs. Distance curve, measured from where cycle 1 reaches zero load to the Adhesive Force.
- Test Requirements: TPA test or a Compression test with the return speed equal to the test speed.

Below are typical Force vs. Distance graphs showing these calculations





Fracture Calculations

A fracture is defined by a sharp drop in load. The sensitivity can be used to specify the minimum required drop, and is expressed as a percent of the Hardness.

Quantity of Fractures

- Sensory Definition: Gives a good indication of food crispiness and crunchiness.
- Mathematical Definition: The number of fractures occurring during the first compression stroke.

- Test Requirements: TPA or Compression test type.

Fracturability

- Sensory Definition: Indicates the brittleness of the food.
- Mathematical Definition: The load value at the first fracture.
- Test Requirements: TPA or Compression test type.

1st Fracture Load Drop Off

- Mathematical Definition: The amount that the load value decreases after the first fracture.
- Test Requirements: TPA or Compression test type.

1st Fracture Work Done

- Mathematical Definition: The area under the Load vs. Distance curve up to the first fracture.
- Test Requirements: TPA or Compression test type.

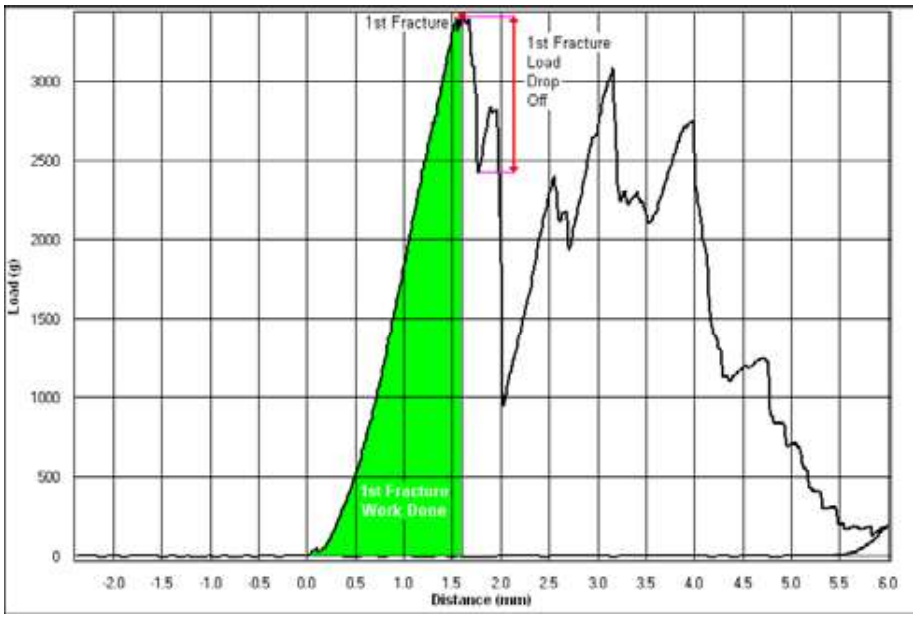
1st Fracture Deformation

- Mathematical Definition: The distance value at the first fracture.
- Test Requirements: TPA or Compression test type.

1st Fracture %Deformation

- Mathematical Definition: Deformation at Hardness / sample length * 100%.
- Test Requirements: TPA or Compression test type; Sample length must be defined.

Below is a typical Load vs. Distance graph showing these calculations.



VI.11 Cohesiveness Calculations

Cohesiveness

- Sensory Definition: Indicates the strength of internal bonds making up the body of the food.
- Mathematical Definition: Hardness Work Done Cycle 2 / Hardness Work Done Cycle 1.
- Test Requirements: TPA test.

Corrected Cohesiveness

- Sensory Definition: Net work invested in the non-recoverable deformations of the first and second bites.
- Mathematical Definition: (Hardness Work Done Cycle 2 - Recoverable Work Done Cycle 2) / (Hardness Work Done Cycle 1 - Recoverable Work Done Cycle 1).
- Test Requirements: TPA test.

VI.12 Springiness Calculations

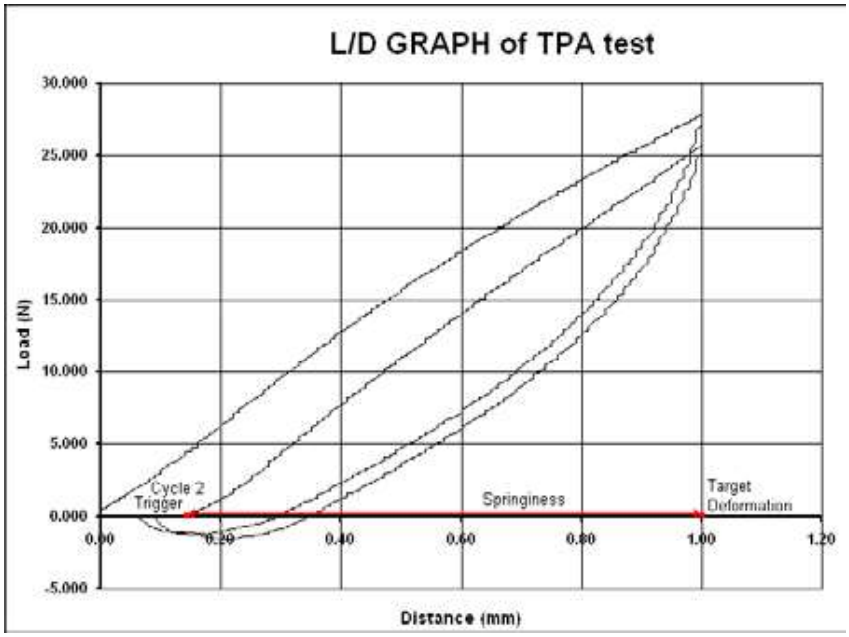
Springiness

- Sensory Definition: Height that the food recovers between the first and the second bites.
- Mathematical Definition: Distance from the target deformation of cycle 1 to the trigger point of cycle 2.
- Test Requirements: TPA test.

Springiness Index

- Sensory Definition: Gives the recovery properties of a food, with a value of 1 indicating a completely elastic material, and a value of 0 indicating a completely viscous material.
- Mathematical Definition: Springiness / distance at target, giving a value between 0 to 1.
- Test Requirements: TPA test.

Below is a typical Force vs. Distance graph showing these calculations.



VI.13 Gumminess Calculations

Gumminess

- Sensory Definition: Indicates the energy required to disintegrate a semi-solid food to a state ready for swallowing. Related to foods with low hardness levels.
- Mathematical Definition: Hardness * Cohesiveness.
- Test Requirements: TPA test.

Corrected Gumminess

- Mathematical Definition: Hardness * Corrected Cohesiveness.
- Test Requirements: TPA test.

VI.14 Chewiness Calculations

Chewiness

- Sensory Definition: Indicates the energy required to chew a solid food product to a state where it is ready for swallowing.
- Mathematical Definition: Hardness * Cohesiveness * Springiness.
- Test Requirements: TPA test.

Chewiness Index

- Mathematical Definition: Hardness * Cohesiveness * Springiness Index.
- Test Requirements: TPA test

Corrected Chewiness

- Mathematical Definition: Hardness * Corrected Cohesiveness * Springiness.
- Test Requirements: TPA test.

Average Peak Load

- Mathematical Definition: The average of the Hardness of each compression cycle.

Mean Load

- Mathematical Definition: The average load value between the specified times.
- Test Requirements: TPA or Compression test type.

VI.15 Tension Calculations

Peak Load

- Sensory Definition: The force required to break a piece of food.
- Mathematical Definition: The maximum load value during the tension test.
- Test Requirements: Tension test.

Deformation at Peak Load

- Sensory Definition: The distance the food sample will stretch before breaking.
- Mathematical Definition: Distance value at the Peak Load.
- Test Requirements: Tension test.

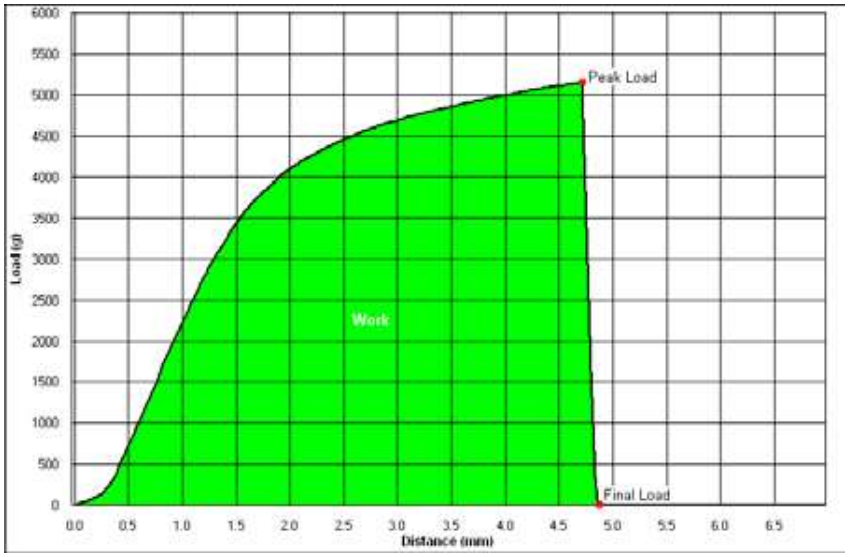
Work

- Sensory Definition: The work necessary to break the food sample.
- Mathematical Definition: Area under the Load vs. Distance curve to the target distance.
- Test Requirements: Tension test.

Final Load

- Sensory Definition: Represents the weight of the upper piece of the broken food sample.
- Mathematical Definition: The load value at the point where the sample breaks.
- Test Requirements: Tension test.

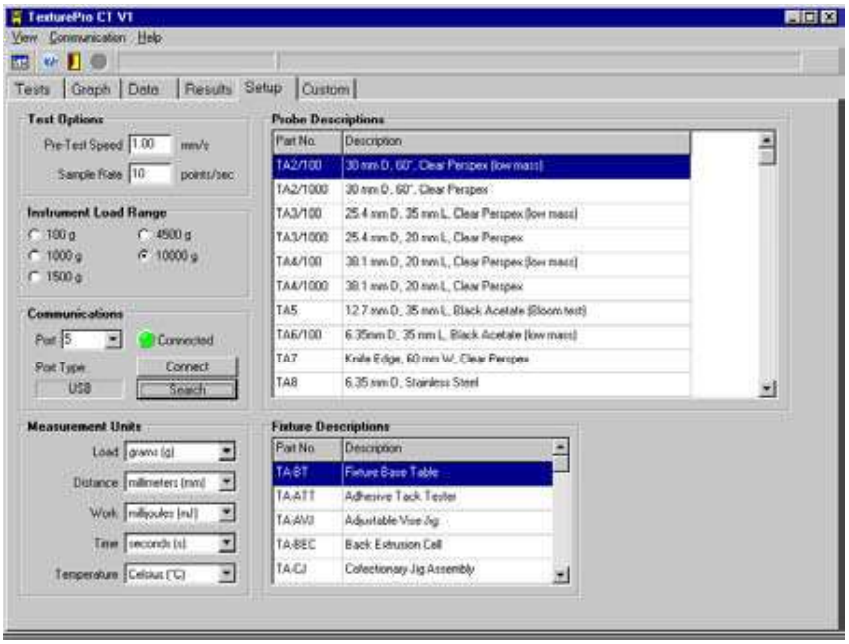
Below is a typical Load vs. Distance graph showing these calculations.



VII. SET UP

VII.1 Setup Page

Use this page to set various program-wide options and to view probe and fixture descriptions.



Choosing a Sample Rate

The Sample Rate determines how often data points are taken during a test: from 10 points/sec to 100 points/sec.

A sample rate of 10 points/sec should be sufficient for most tests and most calculations. If, however, greater detail is desired (say for Fracture or Adhesiveness calculations), the sample rate can be increased.

Note: Increasing the sample rate greatly increases the size of the data file, and will slow down the saving, loading, and graphing of that file

Choosing the Measurement Units

Use the Measurement Unit drop down lists to choose the desired display units for various parameters.

Certain measurement units are determined by other units, or are fixed. Speed units will be the chosen distance units divided by the chosen time units. For example, if centimeters are selected for distance units and minutes for time units, speed will be in centimeters per minute (cm/min). Apparent Modulus and Peak Stress values are always in dynes per square centimeter (dyn/cm^2), and the Sample Rate is always in points per second (points/sec).

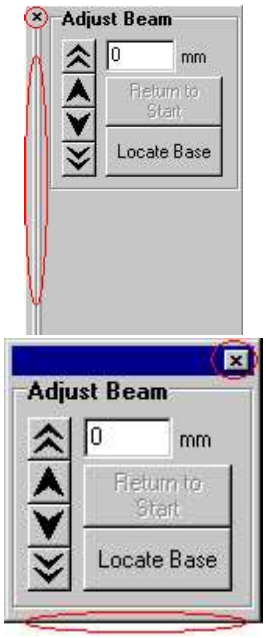
Note: Each time that units are changed, the software must be restarted.

VIII. CUSTOMIZING DISPLAY

VIII.1 Control Windows

Most of the application's controls are grouped together on control windows, which can be moved, resized, or hidden by the user.

- Moving the window: Click and drag the bar on the left of the window.
- Undocking the window: Double-click the bar on the left of the window in order to float it above the application.
- Resizing the window: While the window is undocked, drag on the edges of the window to resize it.
- Hiding the window: Click on the x in the corner of the window.

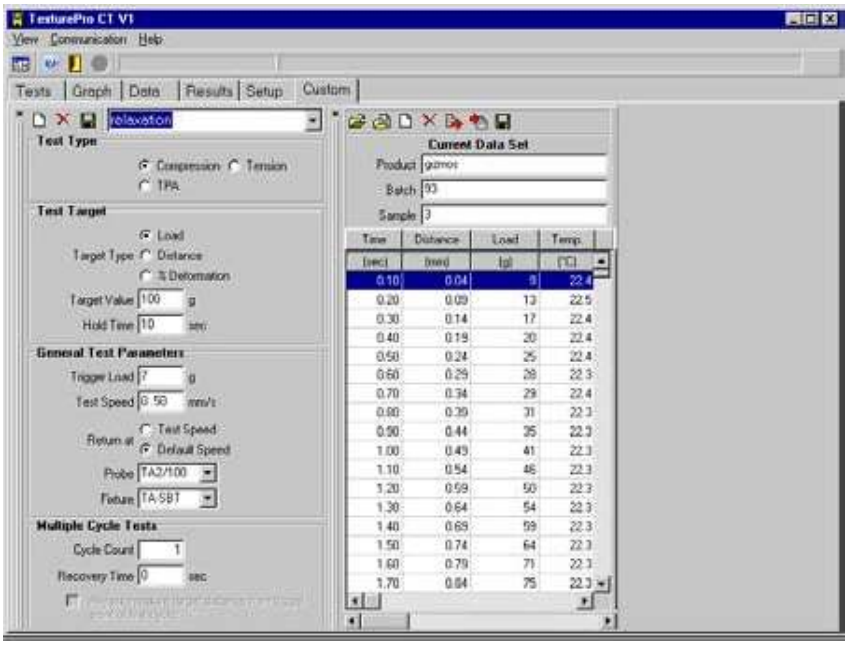


Custom Page

Control windows from other pages can be moved to this page to make a customized display. Control windows can be moved by either:

- Right-clicking on the Custom Page and choosing the control window from the popup menu
- Undocking the control window (by double-clicking the bar on the left of the window) and dragging it to the Custom Page.

An example of a customized display.



View Menu

The View Menu (part of the Main Menu) allows the user to choose which control windows will be visible. Choose the name of a control window from the View menu in order to hide or show that window. This menu is especially useful for finding control windows, which have been hidden.

IX. TROUBLESHOOTING

The "New Hardware" Wizard comes up when the USB cable is connected and instrument is turned on:

1. If the instrument was connected before the software was installed, canceling the wizard and restarting the instrument can now clear this problem.

2. If this problem still occurs after the software is installed, exit the wizard, turn off the instrument, and reinstall the software.

The Instrument goes directly to Standalone mode even though a USB or RS-232 cable is connected.

1. This may happen the very first time the instrument is connected using USB, due to a timing issue: Windows is initialing the USB port while the instrument is looking for it. After restarting the instrument, it should ask if you want to operate in Remote or Standalone mode.

2. If this problem occurs in any other situation, make sure that the cable is securely connected, then restart the instrument.

The USB port is not listed under COM ports on the Setup page.

1. Make sure that the instrument is securely connected, turned on, and in the Remote mode.

2. Click the "Search" button on the Setup page. The software should find the port and reset the instrument.

3. This problem can be avoided by making sure the instrument is connected, turned on, and in the Remote mode before starting the software.

A "General IO Error" is received.

1. Make sure that the instrument is securely connected, turned on, and in the Remote mode.

2. Click either the "Connect" or the "Search" button on the Setup page. The software should reset the instrument.

APPENDIX A - Warranty Repair and Service

Warranty

Brookfield Viscometers are guaranteed for one year from date of purchase against defects in materials and workmanship. The Viscometer must be returned to **Brookfield Engineering Laboratories, Inc.** or the Brookfield dealer from whom it was purchased for no charge warranty evaluation service. Transportation is at the purchaser's expense. The Viscometer should be shipped in its carrying case together with all spindles originally provided with the instrument.

For repair or service in the **United States** return to:

Brookfield Engineering Laboratories, Inc.
11 Commerce Boulevard
Middleboro, MA 02346 U.S.A.

Telephone: (508) 946-6200 FAX: (508) 923-5009
www.brookfieldengineering.com

For repair or service outside the United States consult Brookfield Engineering Laboratories, Inc. or the dealer from whom you purchased the instrument.

For repair or service in the **United Kingdom** return to:

Brookfield Viscometers Limited
1 Whitehall Estate
Flex Meadow, Pinnacles West
Harlow, Essex CM19 5TJ, United Kingdom

Telephone: (44) 27/945 1774 FAX: (44) 27/945 1775
www.brookfield.co.uk

For repair or service in **Germany** return to:

Brookfield Engineering Laboratories Vertriebs GmbH
Hauptstrasse 18
D-73547 Lorch, Germany

Telephone: (49) 7172/927100 FAX: (49) 7172/927105
www.brookfield-gmbh.de

For repair or service in **China** return to:

Guangzhou Brookfield Viscometers and Texture Instruments Service Company Ltd.
Room C1, 5/F, Tianxing Building East Tower, No. 21, Zhongshan Yi Road, Yuexiu District
Guangzhou, 510600, P. R. China

Telephone: (86) 20/3760-0548 FAX: (86) 20/3760-0548
www.brookfield.com.cn

On-site service at your facility is also available from Brookfield. Please contact our Service Department in the United States, United Kingdom, Germany or China for details.